



INTRODUCTION TO DATA COMMUNICATION AND COMPUTER NETWORK (IDCCN) **[CSIT369]**

MODULE III

Physical Layer

(UG - III Sem)

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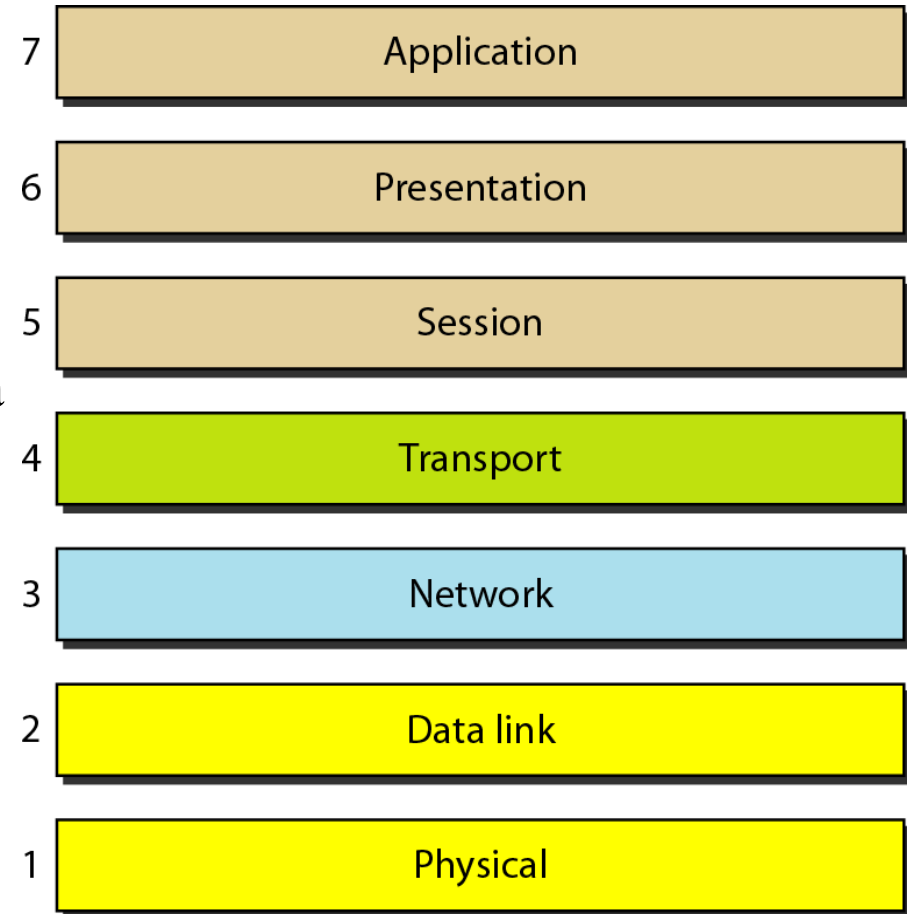
PHYSICAL LAYER

Recap:

- First layer of the OSI Model
- Responsible for transmission of raw bits
- Defines electrical, mechanical, and physical specifications
- Concerned with cables, connectors, signals, and transmission media

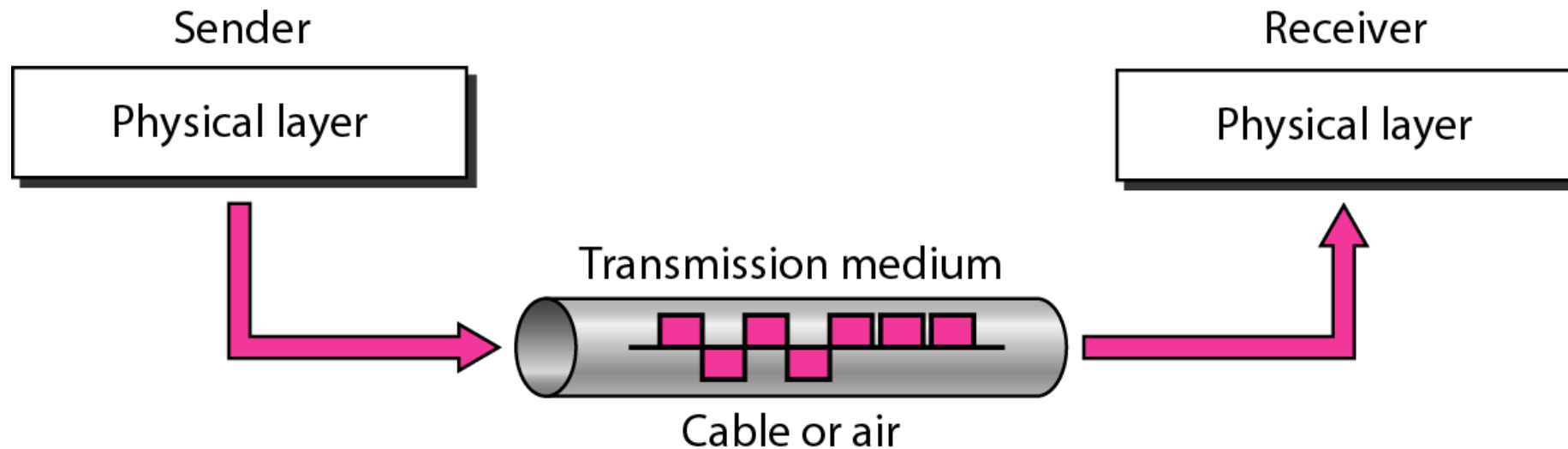
Functions:

- Bit representation
- Data rate control
- Synchronization of bits
- Physical topology
- Transmission media



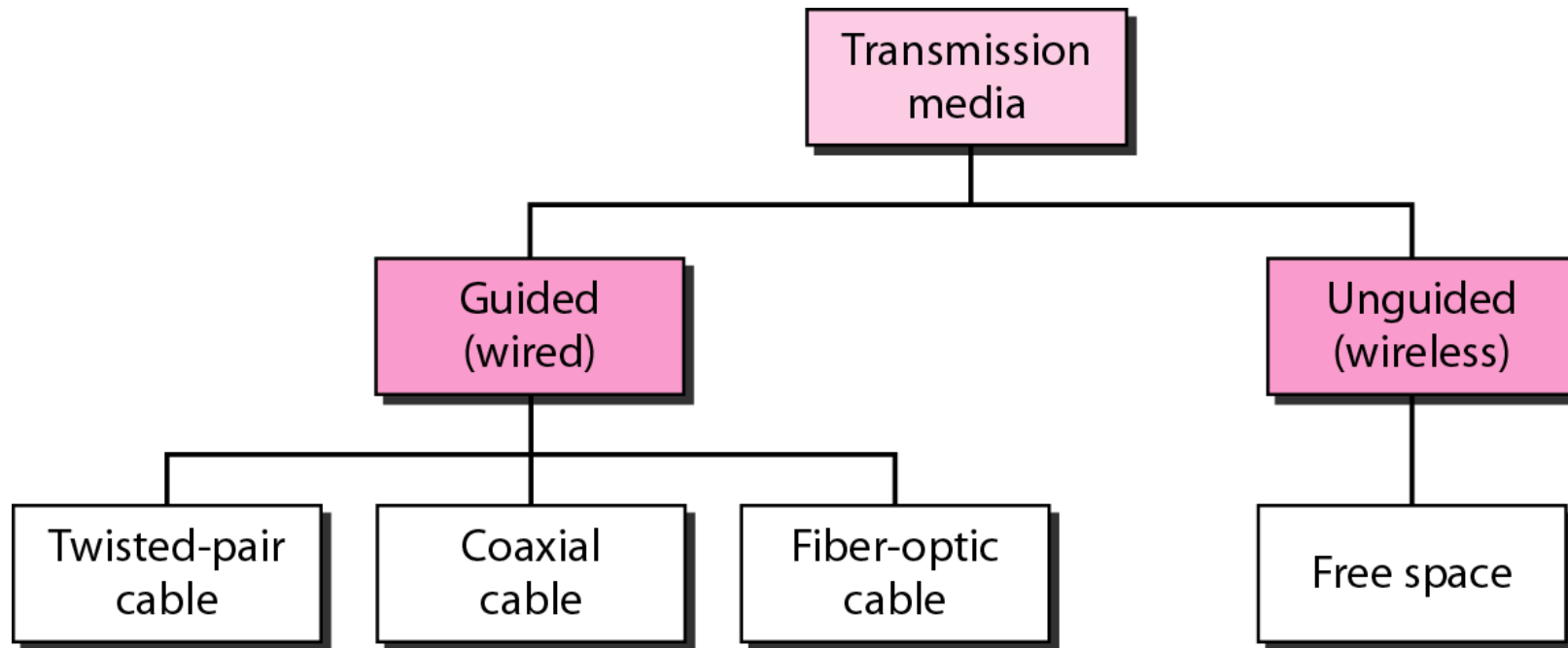
TRANSMISSION MEDIA

- *Communication channel between sender and receiver*





TYPES OF TRANSMISSION MEDIA





GUIDED MEDIA

- ***Guided transmission media**, also known as **wired media**, provide a **physical path (conduit)** that guides data signals from the sender to the receiver.*
- *Data travels through a **solid medium**, such as copper wires or optical fiber, ensuring reliable and secure communication.*

Types of Guided Media:

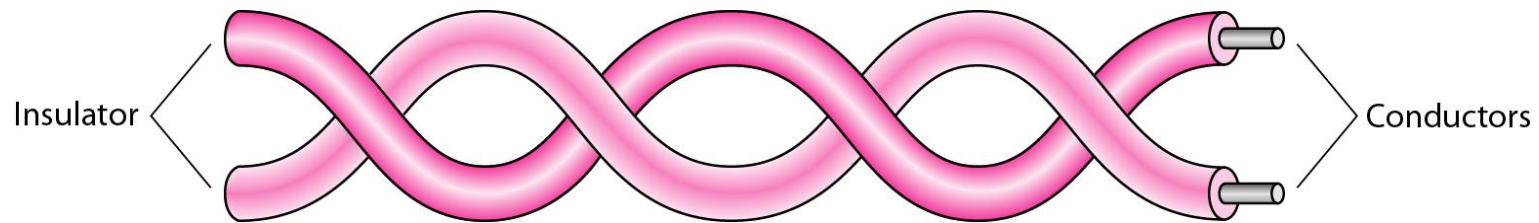
- 1. Twisted Pair Cable**
 1. Unshielded Twisted Pair (UTP)
 2. Shielded Twisted Pair (STP)
- 2. Coaxial Cable**
- 3. Fiber Optic Cable**

TWISTED PAIR CABLE

*A Twisted Pair Cable is the most widely used **guided (wired) transmission medium**, consisting of **two insulated copper wires twisted together** to reduce electromagnetic interference (EMI) and crosstalk.*

UTP (Unshielded Twisted Pair)

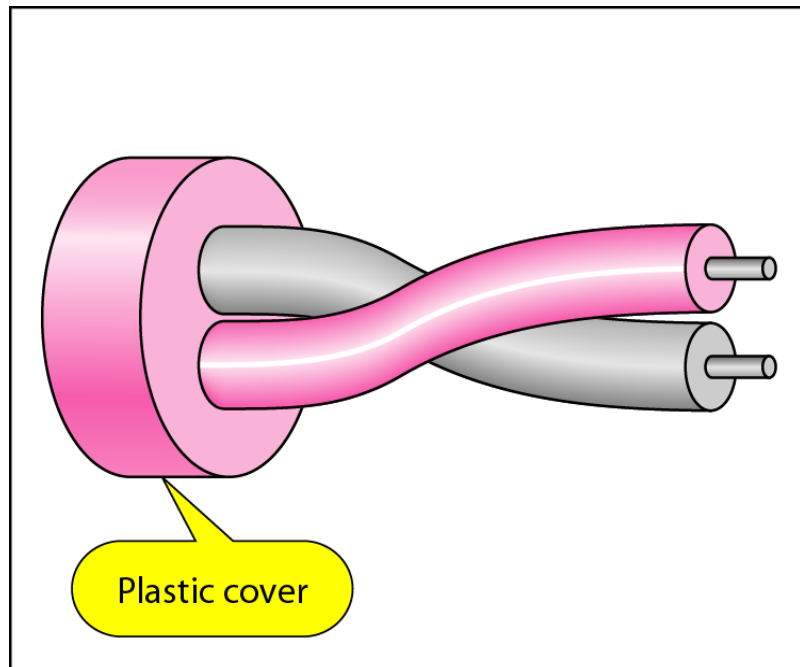
- No additional shielding
- Low cost and easy to install
- Commonly used in Ethernet LANs



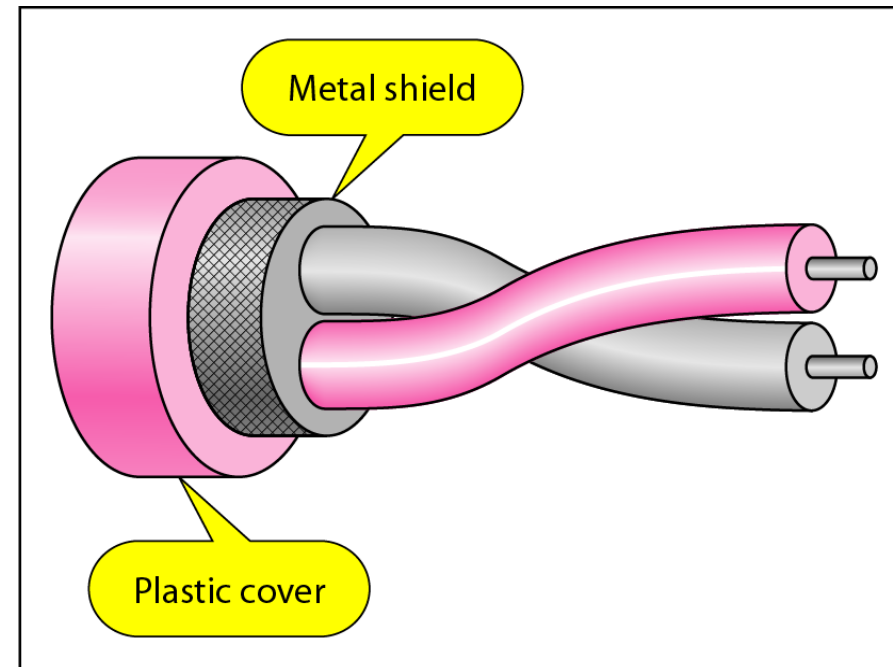
STP (Shielded Twisted Pair)

- Includes a protective metallic shield
- Better protection against electromagnetic interference
- Used in industrial and high-interference environments

UTP AND STP CABLES



a. UTP



b. STP



TWISTED PAIR CABLE

Advantages

- Cost-effective and widely available
- Easy to install and maintain
- Suitable for Local Area Networks (LANs)
- Supports high-speed Ethernet communication

Limitations

- Limited transmission distance (typically up to **100 m** for Ethernet)
- Lower bandwidth than fiber optic cables
- More susceptible to noise than fiber optic cables

Applications

- Telephone networks
- Ethernet (LAN) connections
- Office and home networking



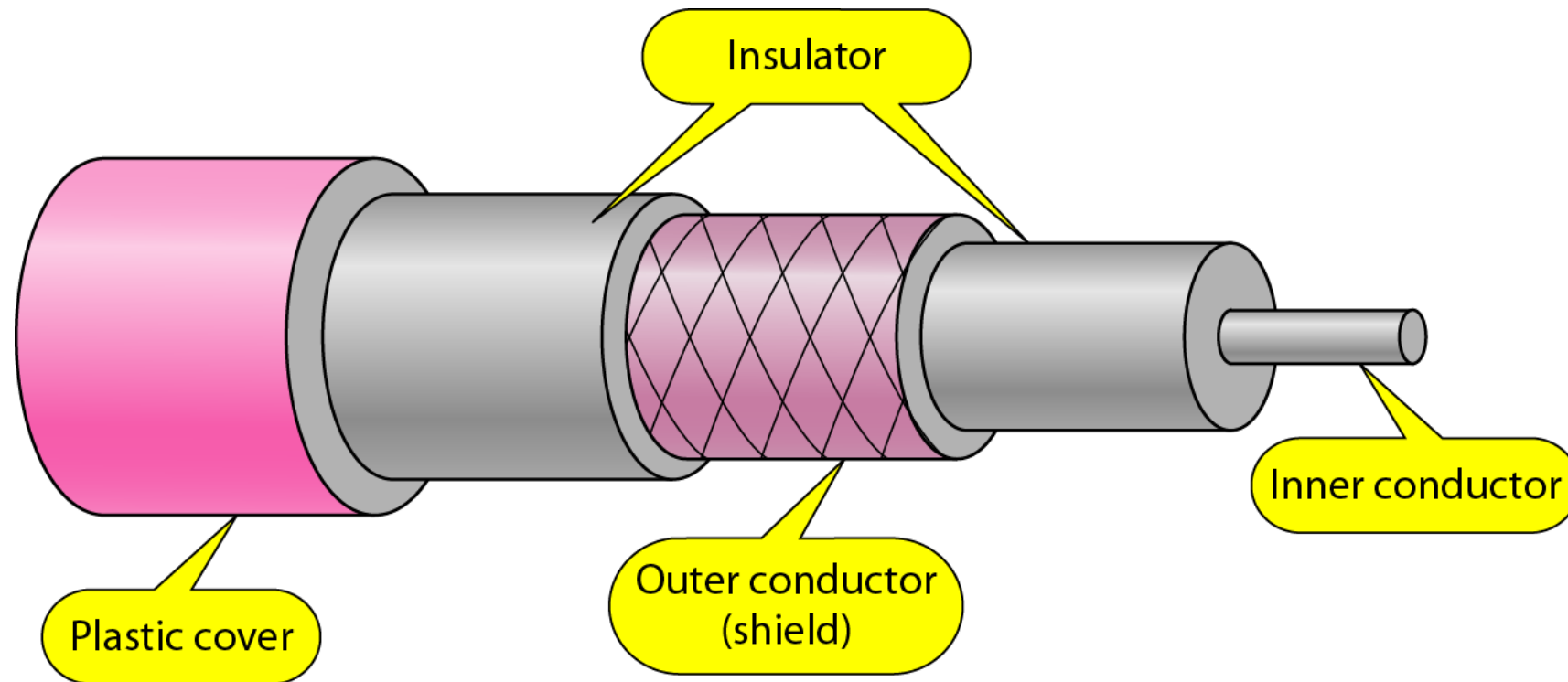
COAXIAL CABLE

- A **Coaxial Cable** is a **guided (wired) transmission medium** that consists of a **inner copper conductor**, an **insulating layer**, a **metallic shield**, and an **outer protective jacket**.
- The metallic shield protects the signal from electromagnetic interference (EMI), enabling reliable data transmission over longer distances than twisted pair cables.

Structure

- **Inner Copper Conductor** – Carries the data signal
- **Insulator** – Separates the conductor from the shield
- **Metallic Shield (Braided/Foil)** – Reduces interference and signal loss
- **Outer Protective Jacket** – Protects the cable from physical damage

COAXIAL CABLE





COAXIAL CABLE

Advantages

- Better resistance to electromagnetic interference (EMI)
- Supports higher bandwidth than twisted pair cables
- Reliable transmission over longer distances
- Durable and robust construction

Limitations

- Bulkier and less flexible than twisted pair cables
- More expensive than twisted pair cables
- Installation and maintenance are relatively difficult

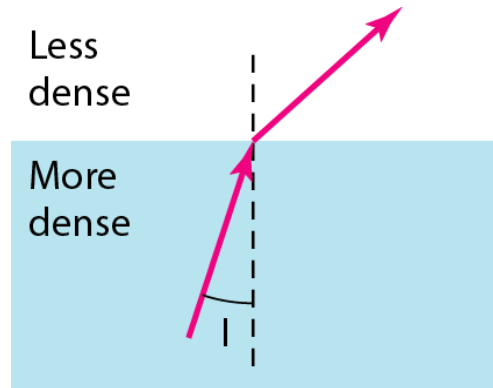
Applications

- Cable Television (CATV) Broadband Internet Connections
- CCTV Surveillance Systems
- Radio Frequency (RF) Signal Transmission

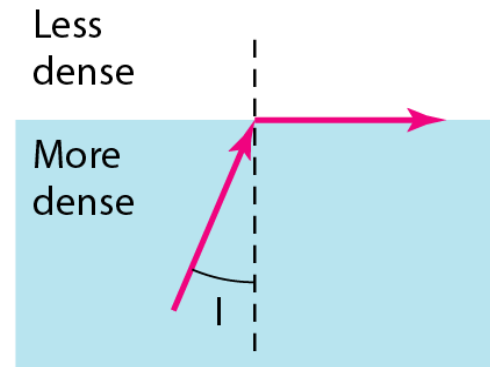
FIBER OPTICS CABLE

- *A Fiber Optic Cable transmits data as light pulses through thin strands of glass or plastic fibers.*
- *It offers very high bandwidth, high-speed communication, and low signal loss, making it ideal for long-distance data transmission.*

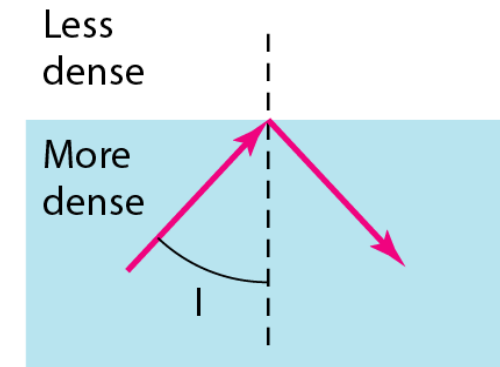
Bending of light ray



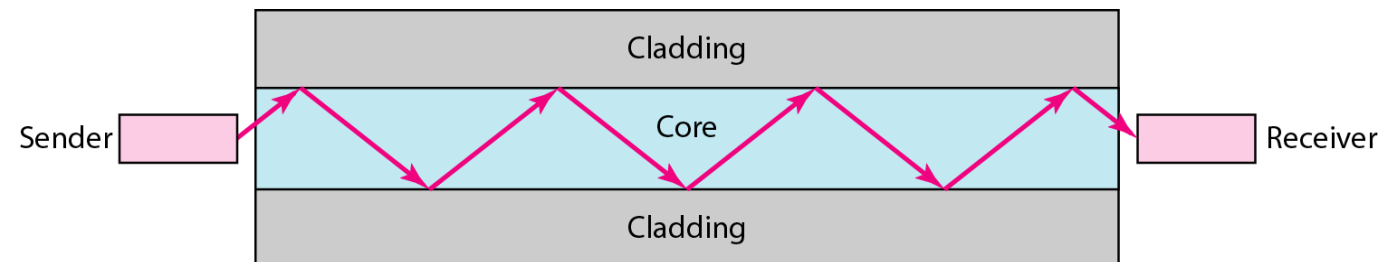
$i < \text{critical angle}$,
refraction



$i = \text{critical angle}$,
refraction



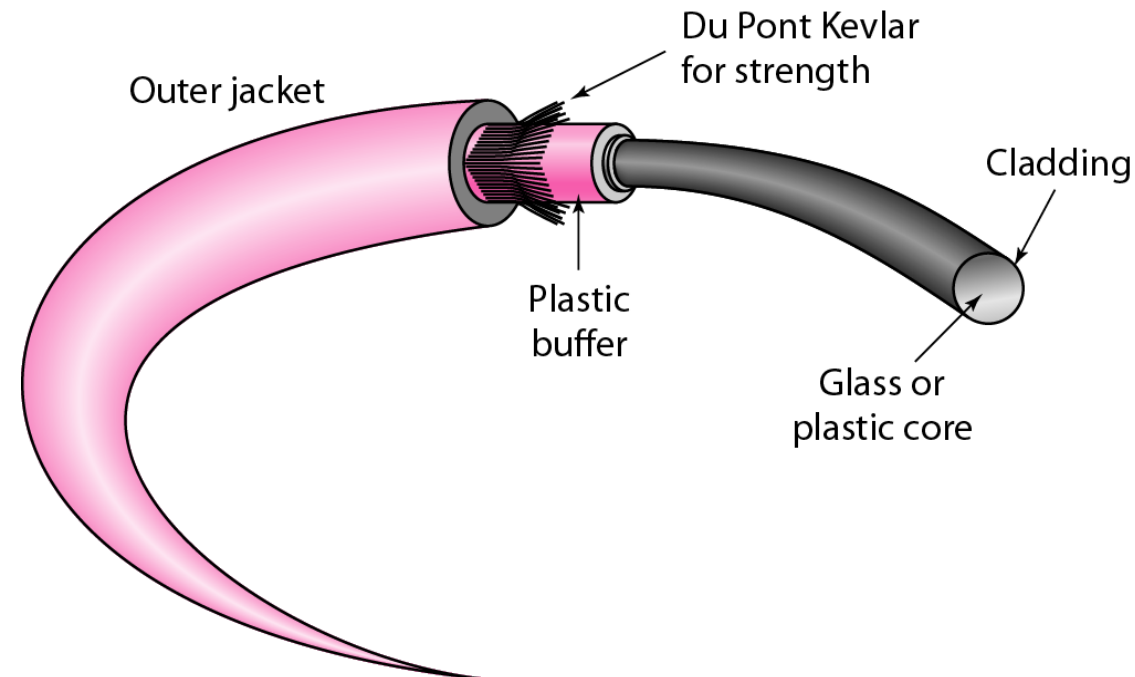
$i > \text{critical angle}$,
reflection



FIBER OPTICS CABLE

Structure

- **Core** – Carries light signals
- **Cladding** – Reflects light back into the core using total internal reflection
- **Buffer Coating** – Protects the fiber from moisture and physical damage
- **Outer Jacket** – Provides mechanical protection





FIBER OPTICS CABLE

Advantages

- Very high data transmission speed
- High bandwidth and long transmission distance
- Immune to electromagnetic interference (EMI)
- Low signal attenuation (loss)
- More secure, as it is difficult to tap

Limitations

- Higher installation and equipment cost
- Fragile compared to copper cables
- Installation and maintenance require specialized tools and skills

Applications

- High-speed Internet and broadband services
- Telecommunication networks
- Data centers and cloud networking
- Cable TV networks



UNGUIDED TRANSMISSION MEDIA (WIRELESS)

Unguided transmission media, also known as wireless media, transmit data through air or free space using electromagnetic waves, without the need for physical cables.

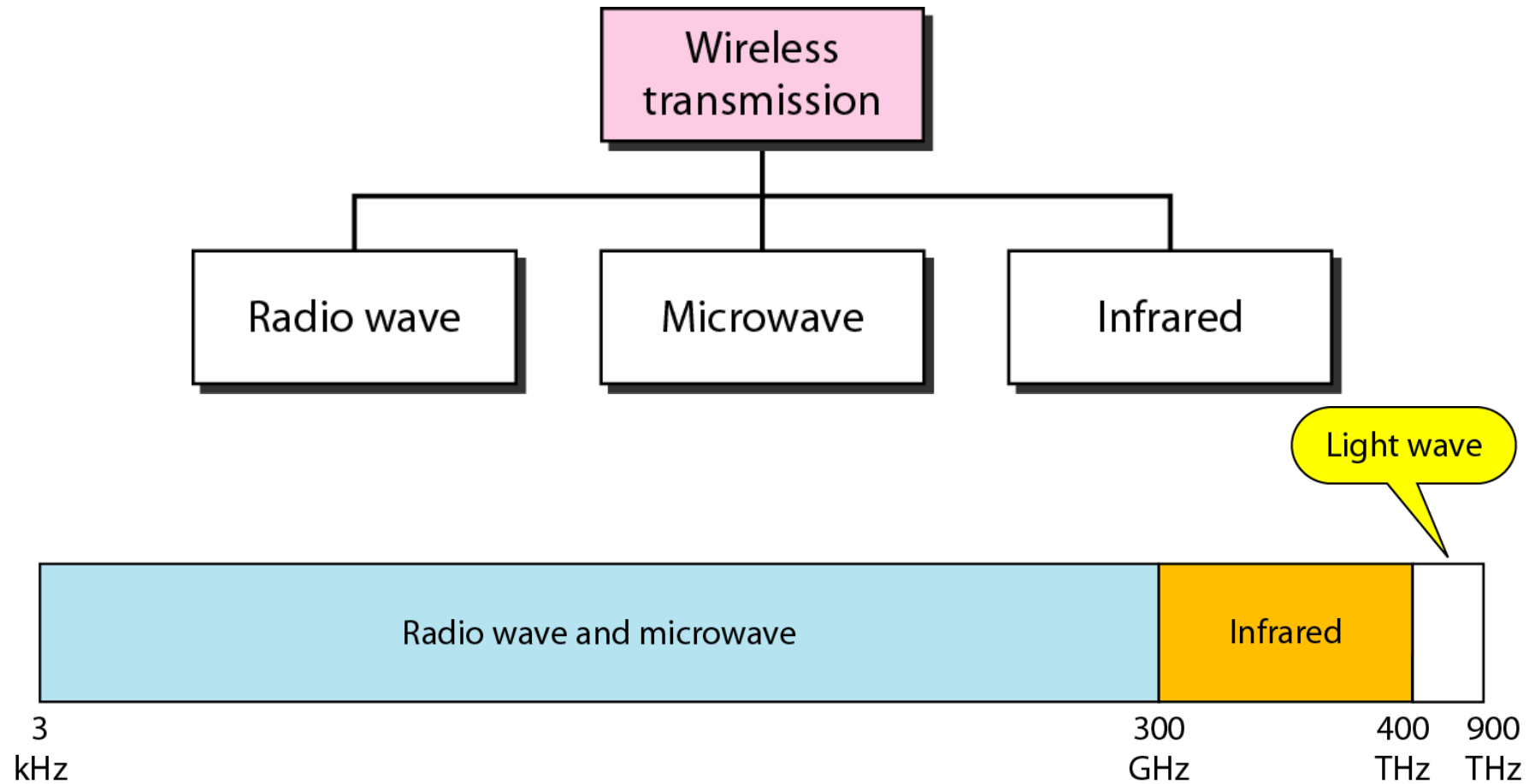
Characteristics

- Uses electromagnetic waves for data transmission.
- Does not require physical cables or connectors.
- Provides mobility and flexibility for users.
- Suitable for both short- and long-distance communication.
- Signal quality may be affected by interference, obstacles, and weather conditions.

Applications

- Wi-Fi Networks and Bluetooth Communication
- Cellular Networks (4G/5G)
- Satellite Communication
- Radio and Television Broadcasting
- Remote Controls and Wireless Peripherals

TYPES OF UNGUIDED TRANSMISSION MEDIA





RADIO WAVES

Characteristics

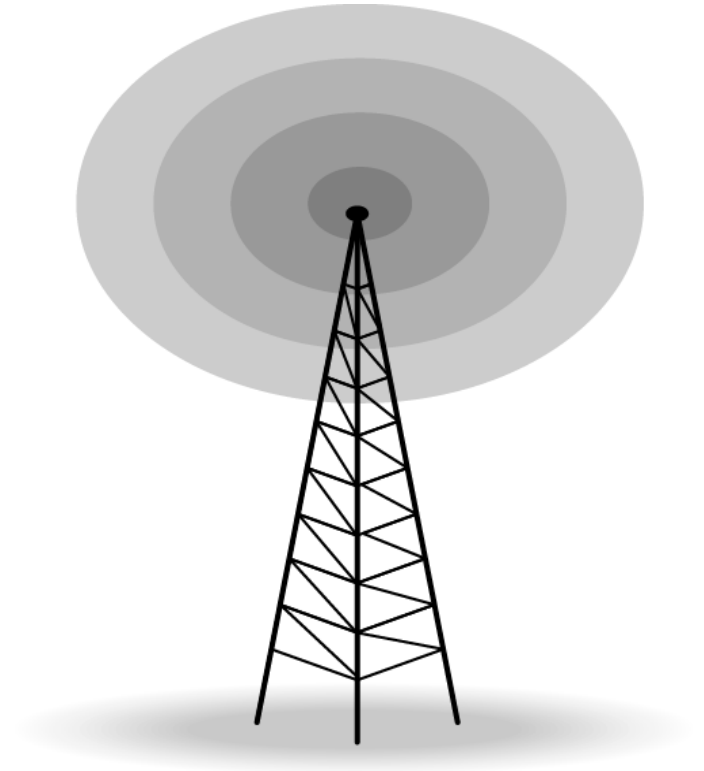
- Frequency Range: **3 kHz – 1 GHz**
- Omnidirectional transmission
- Can penetrate walls and buildings
- Suitable for long-distance communication

Advantages

- Wide coverage area
- Low installation cost
- No direct line of sight required

Applications

- Radio Broadcasting and Television Broadcasting
- Wi-Fi
- Bluetooth





RADIO WAVES

Radio waves are used for multicast communications, such as radio and television, and paging systems.

Radio frequency Allocation:

Radio Spectrum:

- Radio spectrum is the part of the **electromagnetic spectrum** ranging from **1 Hz to 3000 GHz**.
- **Electromagnetic waves in this frequency range, called radio waves**, have become widely used in modern technology, particularly in **telecommunication**.
- The **spectrum is divided into different frequency bands**, and **each band has been allocated** for a specific application ranging from aeronautical, maritime communication, AM and FM radio stations.

Radio frequency Allocation:

- Different parts of the radio spectrum are **allocated** by the **ITU** for different radio transmission technologies and applications.
- In some cases, parts of the **radio spectrum are sold or licensed** to operators of private radio transmission services (example, cellular telephone operators or broadcast television stations). Ranges of allocated frequencies are often referred to by their provisioned use (example, **cellular spectrum** or **television spectrum**).

MICROWAVE

Characteristics

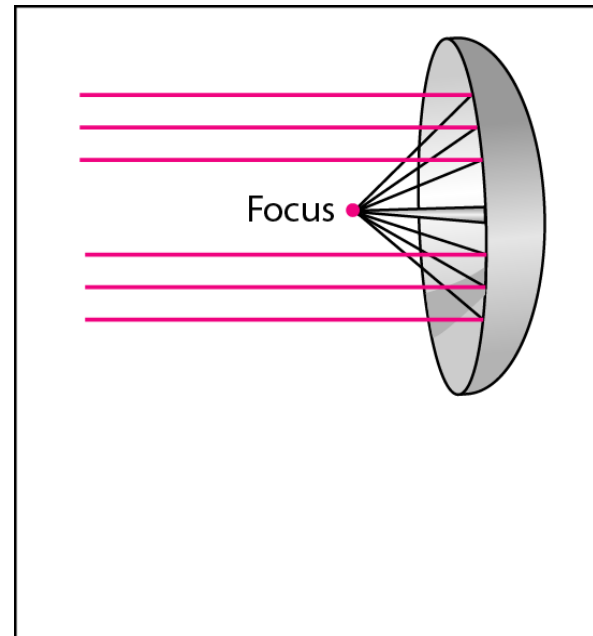
- Frequency Range: **1 GHz – 300 GHz**
- Directional transmission
- High bandwidth
- Requires line-of-sight communication

Advantages

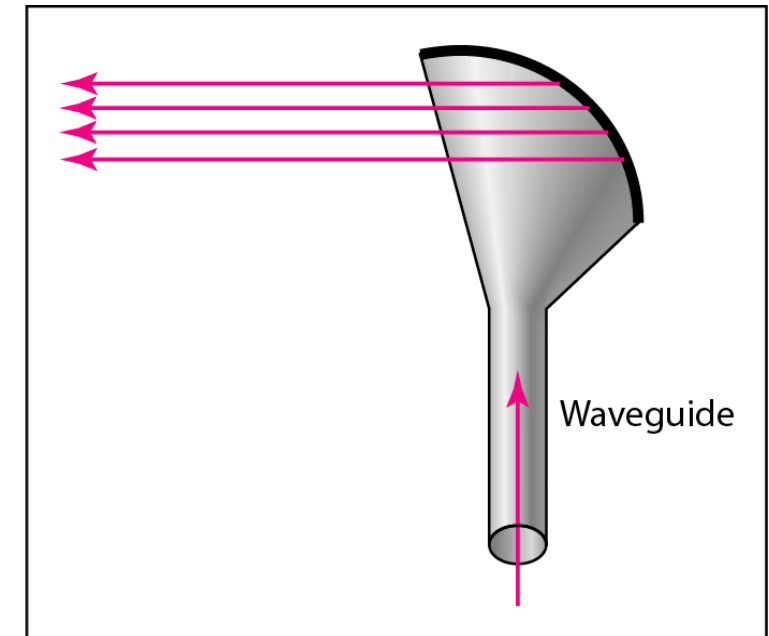
- High-speed data transmission
- Supports large bandwidth
- Suitable for long-distance links

Applications

- Satellite Communication
- Mobile Networks (4G/5G)
- Point-to-Point Communication
- Radar Systems



a. Dish antenna



b. Horn antenna



MICROWAVE

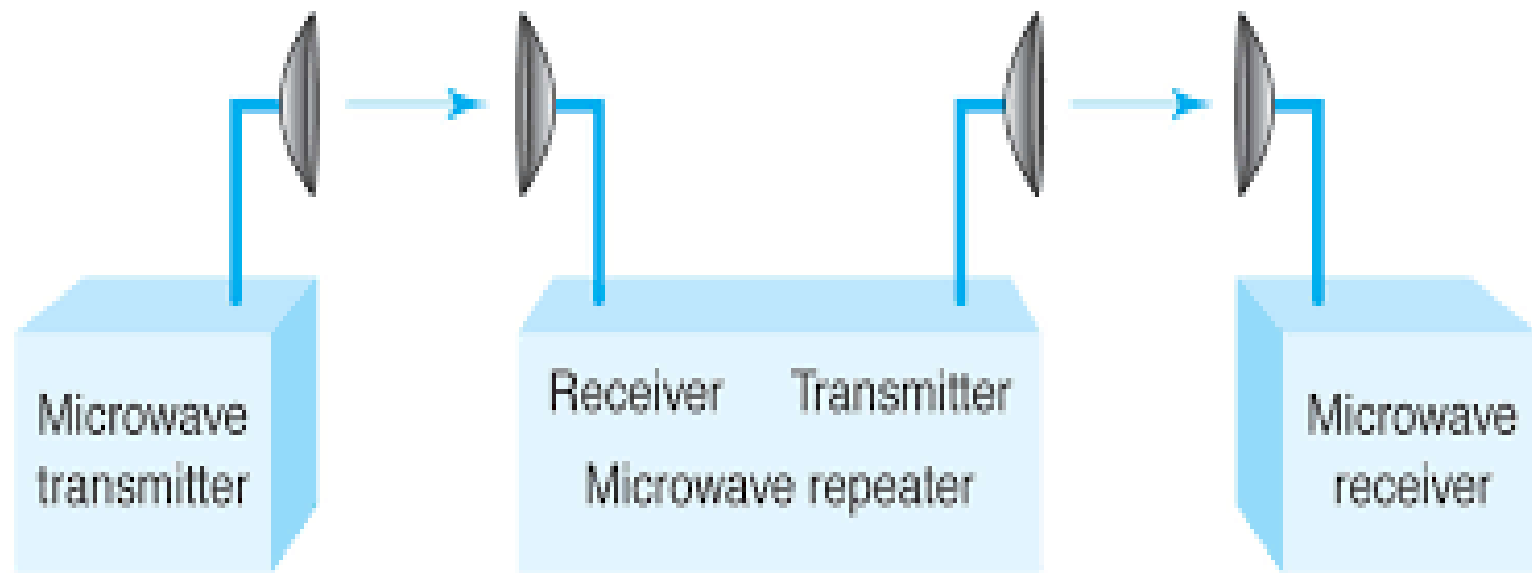
Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

Terrestrial Microwave:

- Electromagnetic waves having **frequencies** between 1 and 300 GHz are called microwaves.
- Microwaves are **unidirectional**; when an antenna transmits microwaves, they can be narrowly focused. This means that the **sending and receiving antennas need to be aligned**. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.
- Microwave's propagation is **Line-of-sight (LOS)**. Since the towers with the **mounted antennas** needs to be in direct sight of each other, towers that are far apart need to be very tall, the **curvature of the earth** as well as other blocking obstacles do not allow two short towers to communicate using microwaves.
- **Repeaters** are often needed for long distance communication very high frequency microwaves cannot penetrate walls. Example: **satellite**
- Parabolic dish antenna and horn antenna are used for this means of transmission.
- **A magnetron generates microwaves in the form of radio signals.**

- **Line-of-sight (LOS):** A clear path-free of any obstructions between points of microwave signal transmission and reception.
- **Link:** The connection of two fixed microwave sites via a line-of-sight (LOS) path. Also referred to as a “hop.”
- The characteristics of the antennas used in point-to-point communication allow the **same frequencies** to be used throughout a system- that is, the system can employ high frequency reuse.







INFRARED

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.



Television



Infrared Radiations



Remote



INFRARED

Characteristics

- Frequency Range: **300 GHz – 400 THz**
- Short communication range
- Cannot pass through solid objects
- Secure due to limited coverage
- Low power consumption

Applications

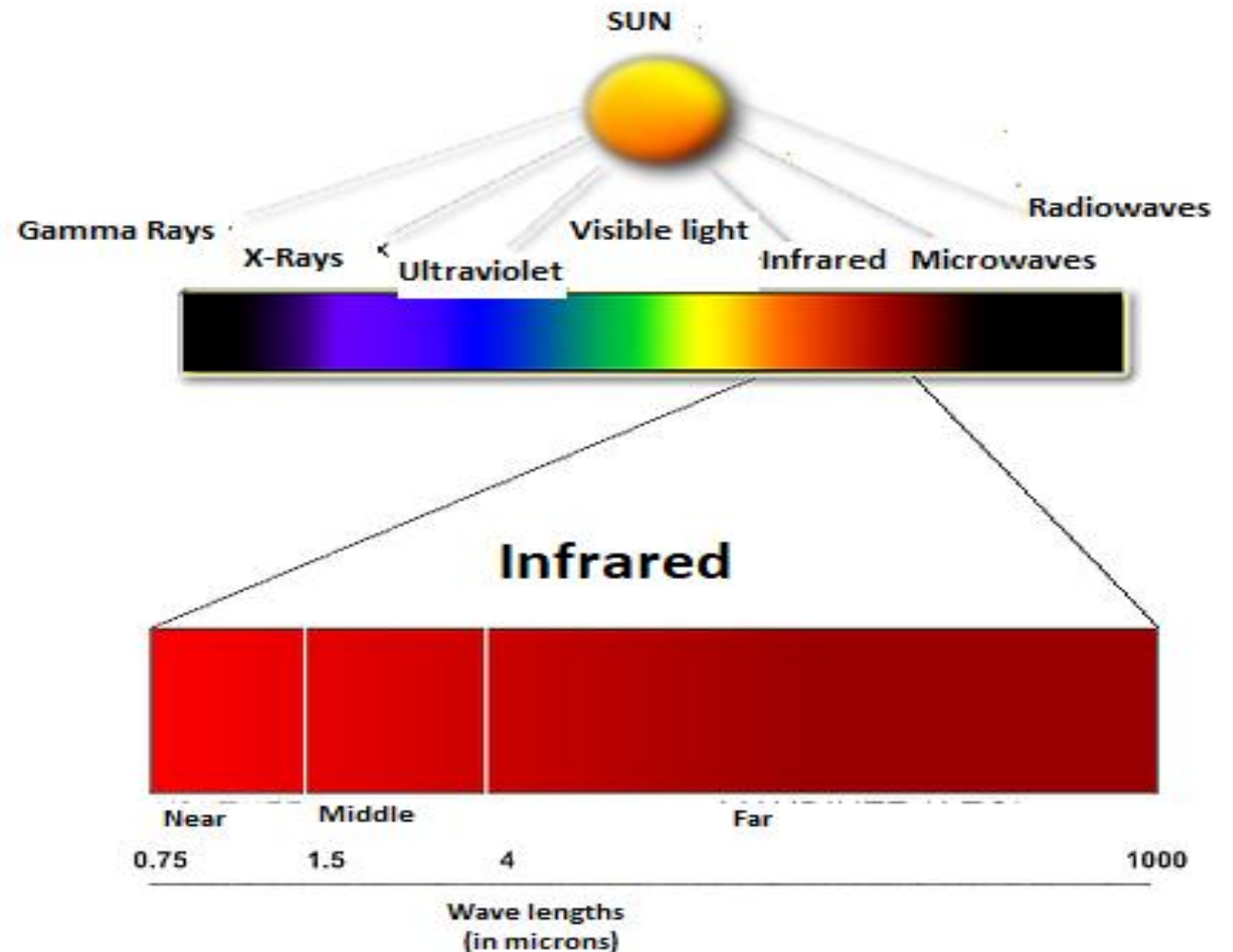
- TV Remote Controls
- Wireless Keyboard and Mouse
- Short-Range Device Communication
- Infrared Sensors

Infrared rays:

- Infrared signals with frequencies ranges from 300 GHz to 400 GHz can be used for short range communication.
- Infrared signals, having **high frequencies, cannot penetrate walls**. This helps to prevent interference between one system and another. Infrared Transmission in one room cannot be affected by the infrared transmission in another room.
- Infrared band, has an excellent potential for data transmission. Transfer digital data is possible with a high speed with a very high frequency.
- There are number of computer devices which are used to send the data through infrared medium e.g., keyboard mice, PCs and printers. There are some manufacturers provide a special part called the IrDA (Infrared Data Association) port that allows a wireless keyboard to communicate with a PC.
- They cannot penetrate walls and sun's infrared rays interfere with these rays. So cannot be used for long -range communication.

Examples:

- Home-entertainment remote-control boxes
- Wireless (local area networks)
- Links between notebook computers and desktop computers
- Cordless modem
- Intrusion detectors
- Motion detectors
- Fire sensors
- Night-vision systems
- Medical diagnostic equipment
- Missile guidance systems
- Geological monitoring devices



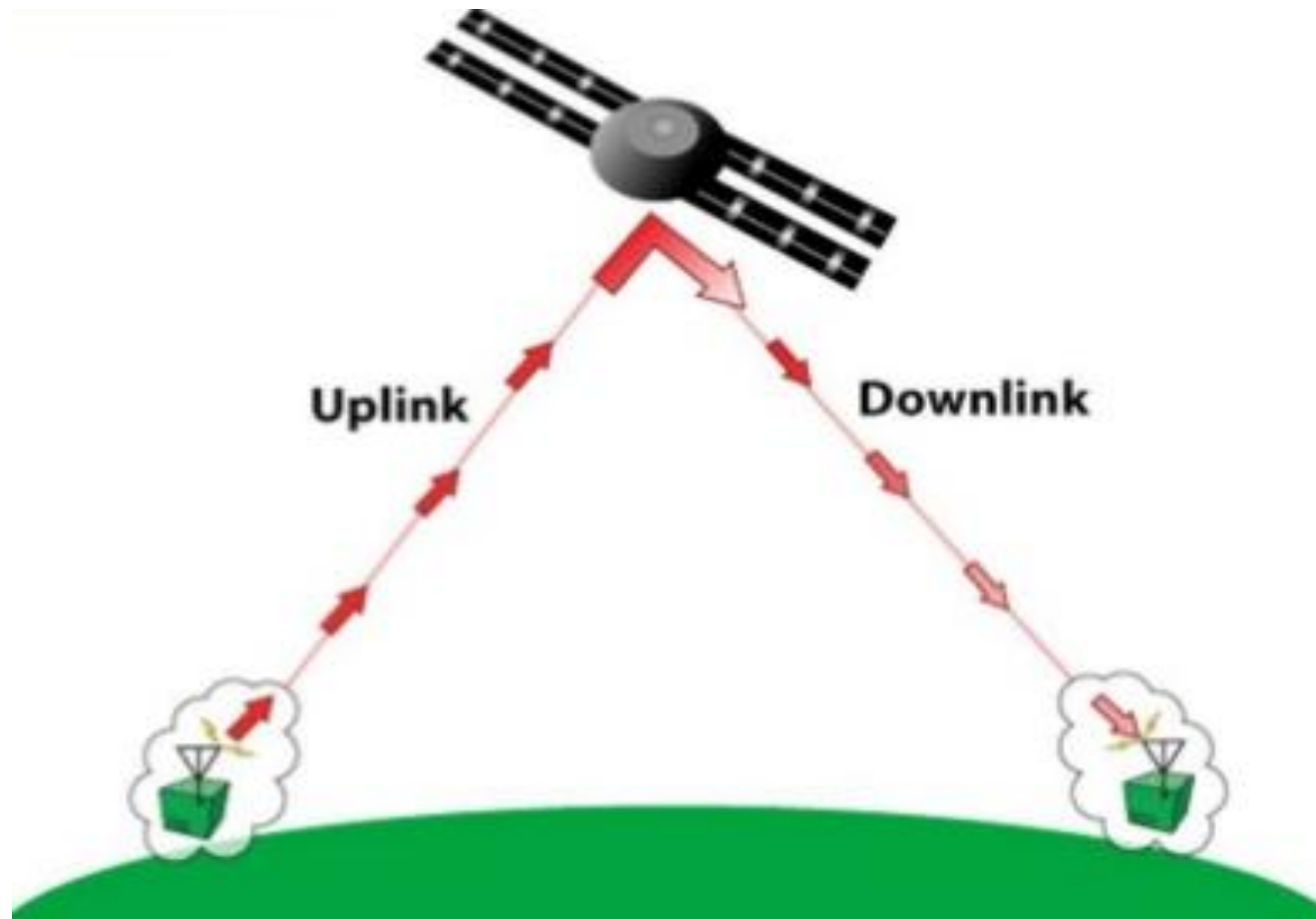
Satellite Communication:

- A satellite is a **space vehicle** launched by humans and orbits Earth or another celestial body.
- **Satellites orbit around the earth.** Depending on the application, these orbits can be circular or elliptical.
- A communication satellite is a station in space that is used for **telecommunication, radio** and **television** signals.
- Satellites are specifically made for telecommunication purpose. They are used for **mobile** applications such as communication to **ships, vehicles, planes, hand-held terminals**, TV and radio broadcasting.
- A satellite works most efficiently when the transmissions are focused with a desired area. When the area is focused, then the emissions don't go outside that designated area and thus minimizing the interference to the other systems. This leads more efficient spectrum usage.

- A communications satellite is a **microwave repeater** in the sky that consists of **Receiver, Transmitter, Amplifier / Regenerator**, Multiplexer, demultiplexer etc.
- Transponder listens to some portion of spectrum, amplifies the incoming signal and broadcasts it in another frequency to avoid interference with incoming signals.



- The **two stations** can use a satellite as a relay station for their communication
- **One Earth Station sends** a transmission to the satellite (This is called an **Uplink**)
- The satellite **Transponder** converts the signal and **sends it down** to the second earth station (This is called a **Downlink**)

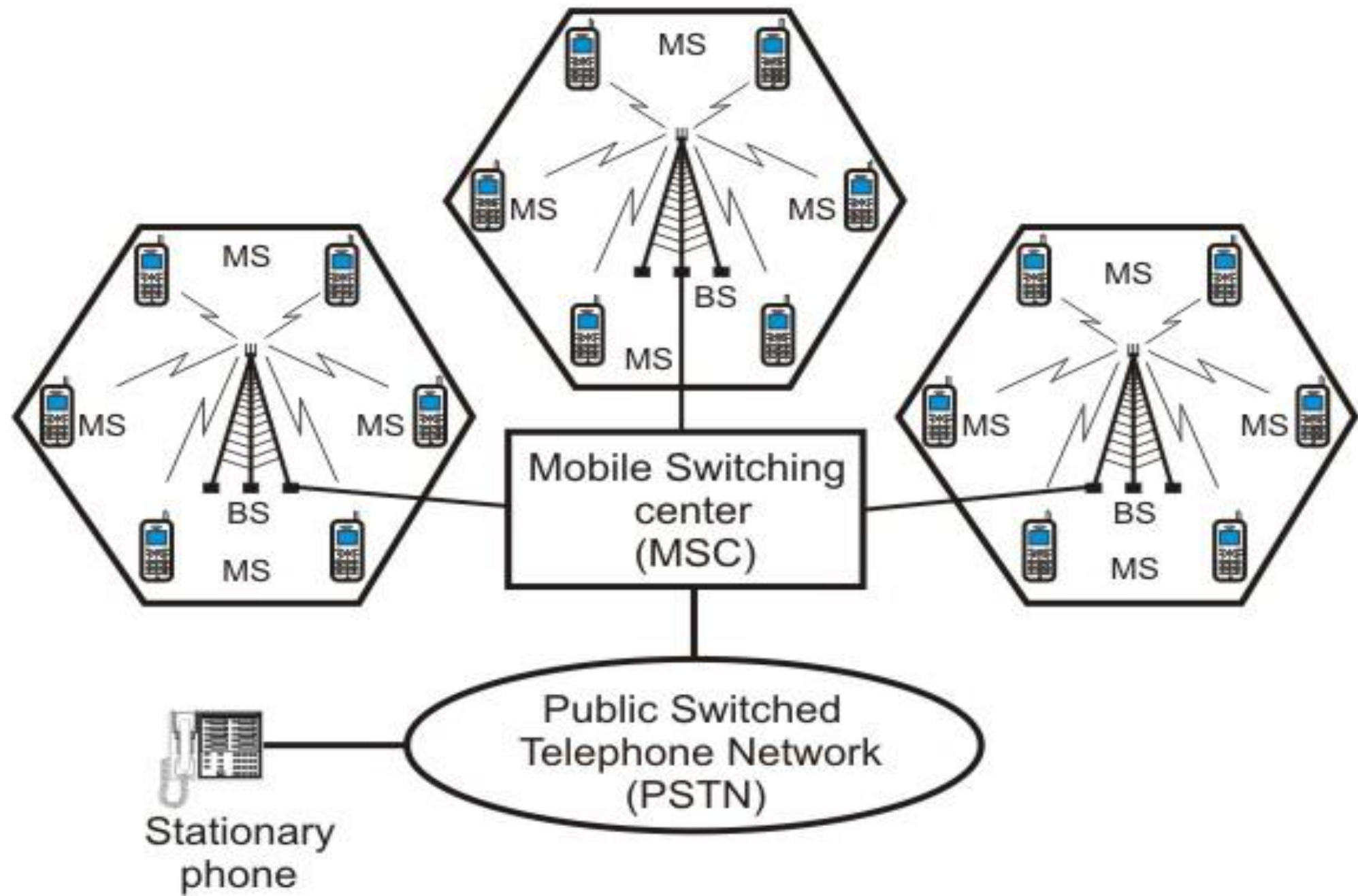


Applications of Satellites communication:

- o Weather Forecasting
- o Radio and TV Broadcast
- o Military
- o Navigation
- o Global Telephone
- o Connecting Remote Areas
- o Global Mobile Communication

Cellular Telephony:

- Cellular telephony is designed to provide **communications** between **two moving units**, called mobile stations (MS), or between one mobile unit and one stationary unit, often called a land unit.
- To accommodate many users over a large geographic area, the cellular telephone system uses many low-power wireless transmitters to create **cells**.
- Variable power levels allow cells to be sized according to subscriber **density** and **demand** within a particular region.
- As **mobile users travel from cell to cell**, their conversations are handed off between cells.
- **Mobile Stations (MS)**: Mobile handsets, which is used by a user to communicate with another user.
- **Cell**: Each cellular service area is divided into small regions called cell (5 to 20 Km).
- **Base Stations (BS)**: Each cell contains an antenna, which is controlled by a small office.
- **Mobile Switching Center (MSC)**: Each base station is controlled by a switching office, called mobile switching center.



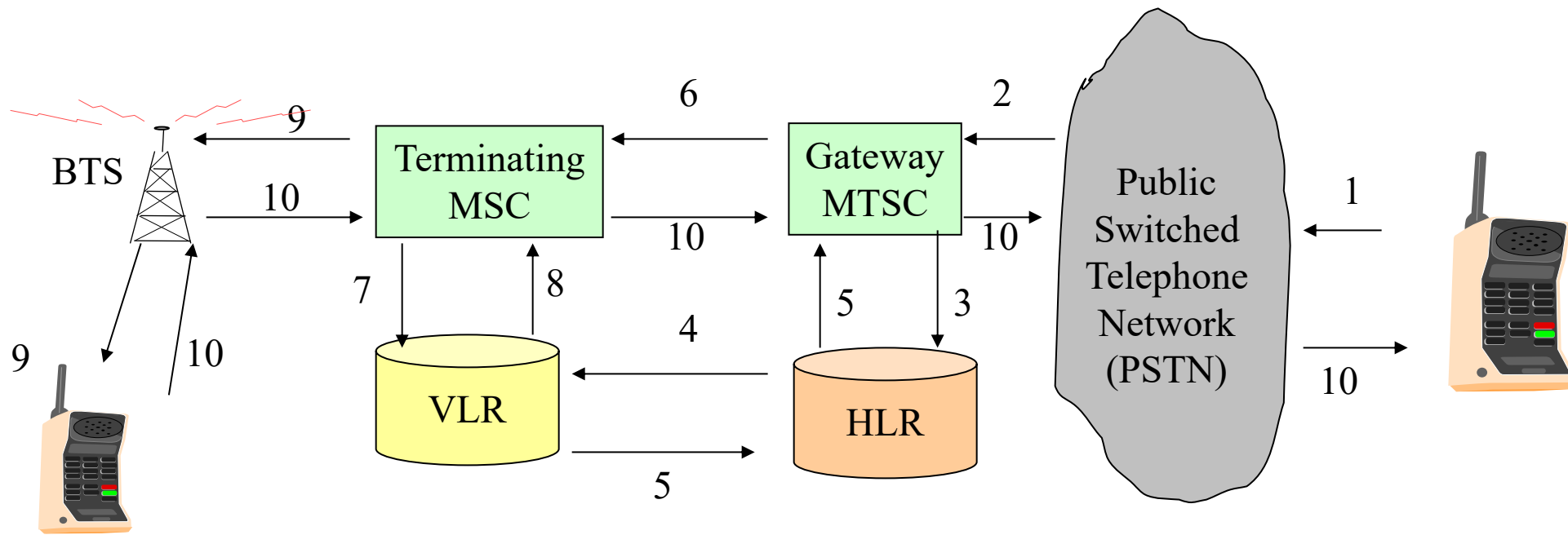
Transmitting and Receiving:

Transmitting involves the following steps:

- A caller enters a 9 / 10-digit code (phone number) and presses the send button.
- The MS scans the band to select a free channel and sends a strong signal to send the number entered.
- The BS relays the number to the MSC.
- The MSC in turn dispatches the request to all the base stations in the cellular system.
- The Mobile Identification Number (MIN) is then broadcast over all the forward control channels throughout the cellular system. It is known as paging.
- The MS responds by identifying itself over the reverse control channel.
- The BS relays the acknowledgement sent by the mobile and informs the MSC about the handshake.
- The MSC assigns an unused voice channel to the call and call is established.

Receiving involves the following steps:

- All the idle mobile stations continuously listens to the paging signal to detect messages directed at them.
- When a call is placed to a mobile station, a packet is sent to the callee's home MSC to find out where it is.
- A packet is sent to the base station in its current cell, which then sends a broadcast on the paging channel.
- The callee MS responds on the control channel.
- In response, a voice channel is assigned, and ringing starts at the MS.



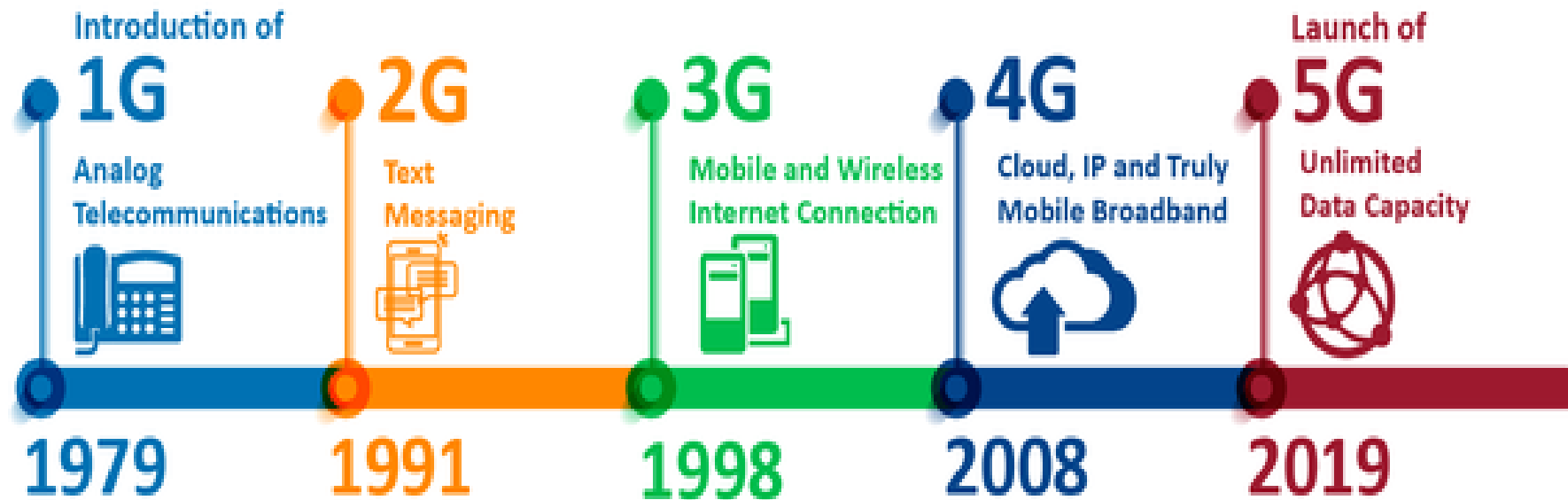
1. Call made to mobile unit (cellular phone)
2. Telephone network recognizes number and gives to gateway MTSC
3. MSC can't route further, interrogates user's HLR
4. Interrogates VLR currently serving user (roaming number request)
5. Routing number returned to HLR and then to gateway MSC

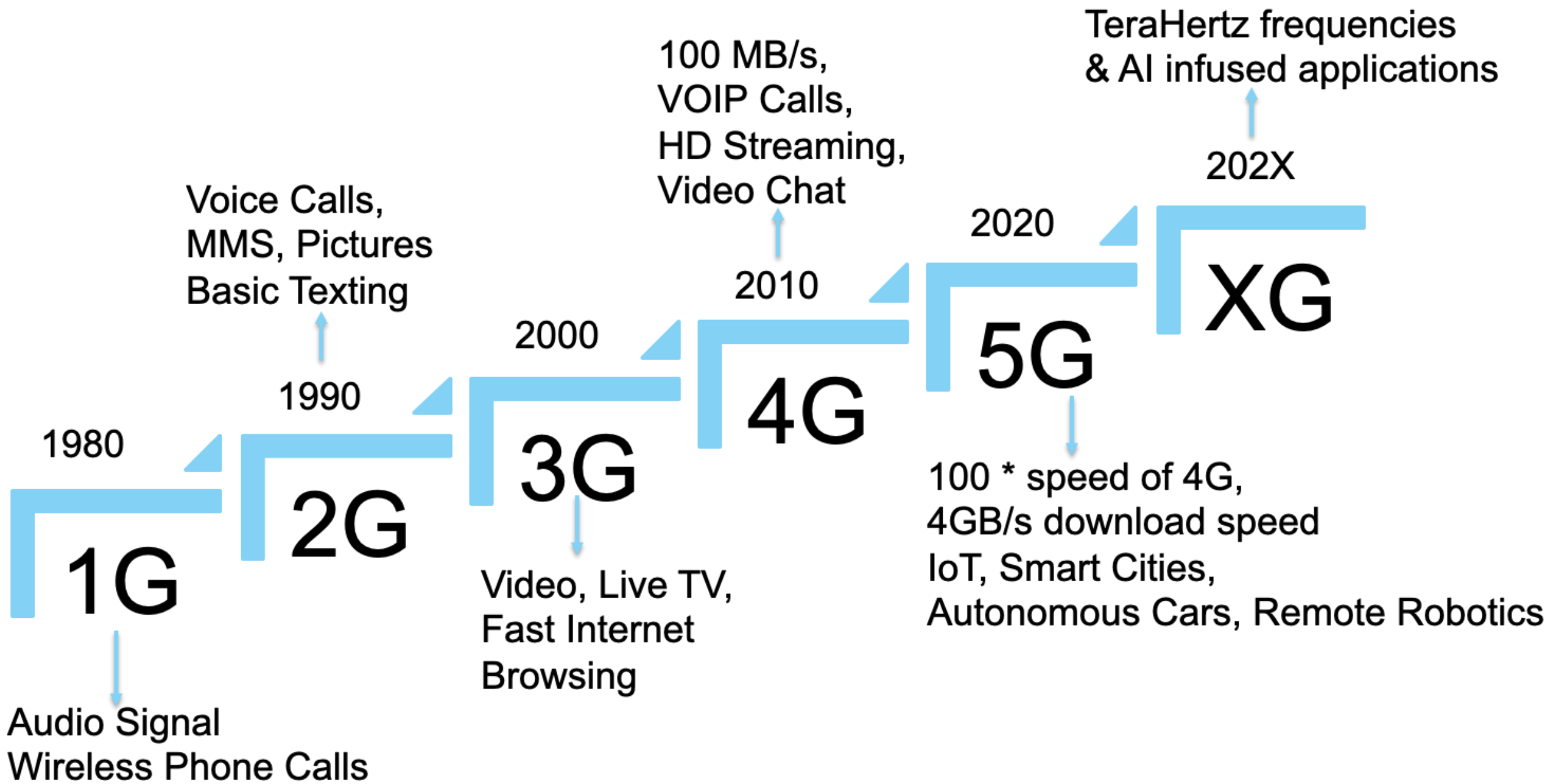
6. Call routed to terminating MSC
7. MSC asks VLR to correlate call to the subscriber
8. VLR complies
9. Mobile unit is paged
10. Mobile unit responds, MSCs convey information back to telephone

MTSC= Mobile Telephone Service Center, **BTS** = Base Transceiver Station

HLR=Home Location Register, **VLR**=Visiting Location Register

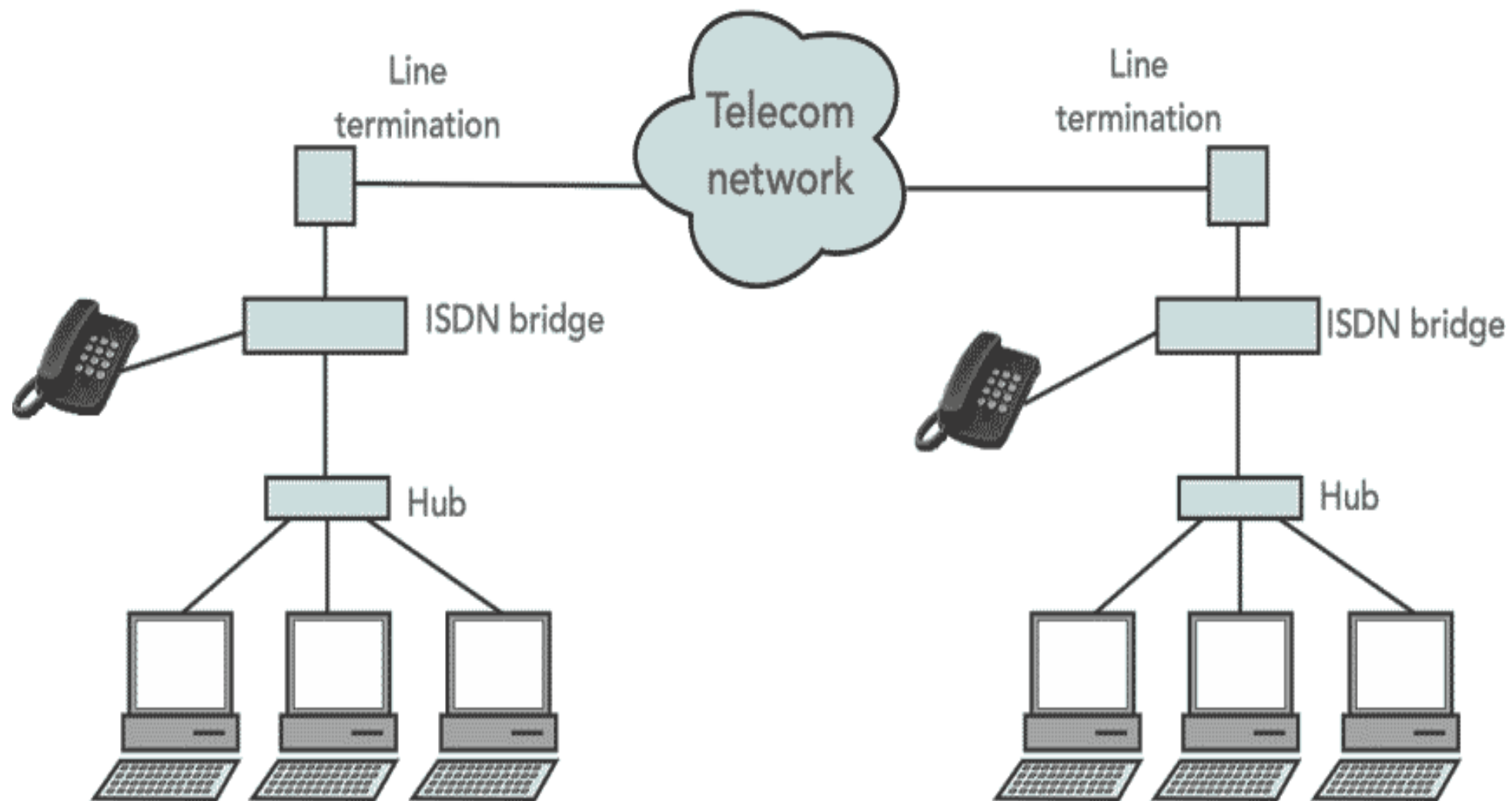
The Evolution of 5G





Introduction to ISDN (Integrated Services Digital Network):

- Earlier, the transmission of **data and voice both** were possible through normal Plain Old Telephone Systems (**POTS**).
- The **sending and receiving of data along with voice was not an easy task**. One could use either the **Internet or the Telephone**. The invention of ISDN helped mitigate this problem.
- The process of connecting a **home computer to the Internet Service Provider** used to take a lot of effort. The usage of the modulator-demodulator unit, simply called the **MODEM** was the essential thing to establish a connection.
- As the system must use the **telephone cable through the telephone exchange for using the Internet**, the **usage of telephone for voice calls was not permitted**. The introduction of ISDN has resolved this problem allowing the **transmission of both voice and data simultaneously**.
- ISDN is a **telephone network-based infrastructure** that allows the transmission of **voice and data simultaneously** at a high speed with greater efficiency.



ISDN supports a variety of services as below:

- Voice calls
- Facsimile
- Electronic Mail
- Database access
- Data transmission and voice
- Connection to internet
- Electronic Fund transfer
- Image and graphics exchange
- Document storage and transfer
- Audio and Video Conferencing
- Automatic alarm services to fire stations, police, medical etc.

Advantages of ISDN:

- ISDN is a telephone network-based infrastructure, which enables the transmission of both voice and data simultaneously.

There are many advantages of ISDN such as

- As the services are digital, there is less chance for errors.
- The connection is faster.
- The bandwidth is higher.
- Voice, data and video – all of these can be sent over a single ISDN line.

Disadvantages of ISDN:

- The disadvantage of ISDN is that it requires specialized digital services and is costlier.

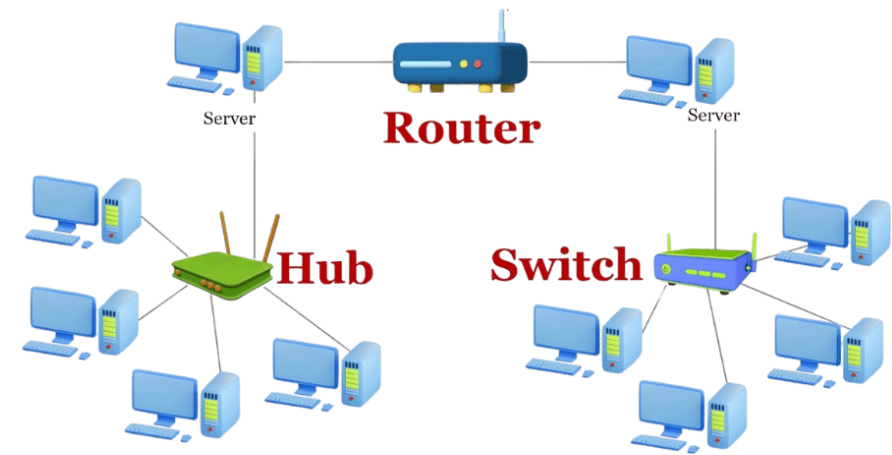
NETWORKING DEVICES

Real-Life Scenario: Imagine sending a message from your laptop to a friend in another city.

- How does the data leave your computer?
- Through which medium does it travel?
- How is it forwarded to the correct destination?

Networking Solution: Physical media and networking devices work together to:

- Connect devices
- Transfer data
- Extend network coverage
- Direct traffic efficiently





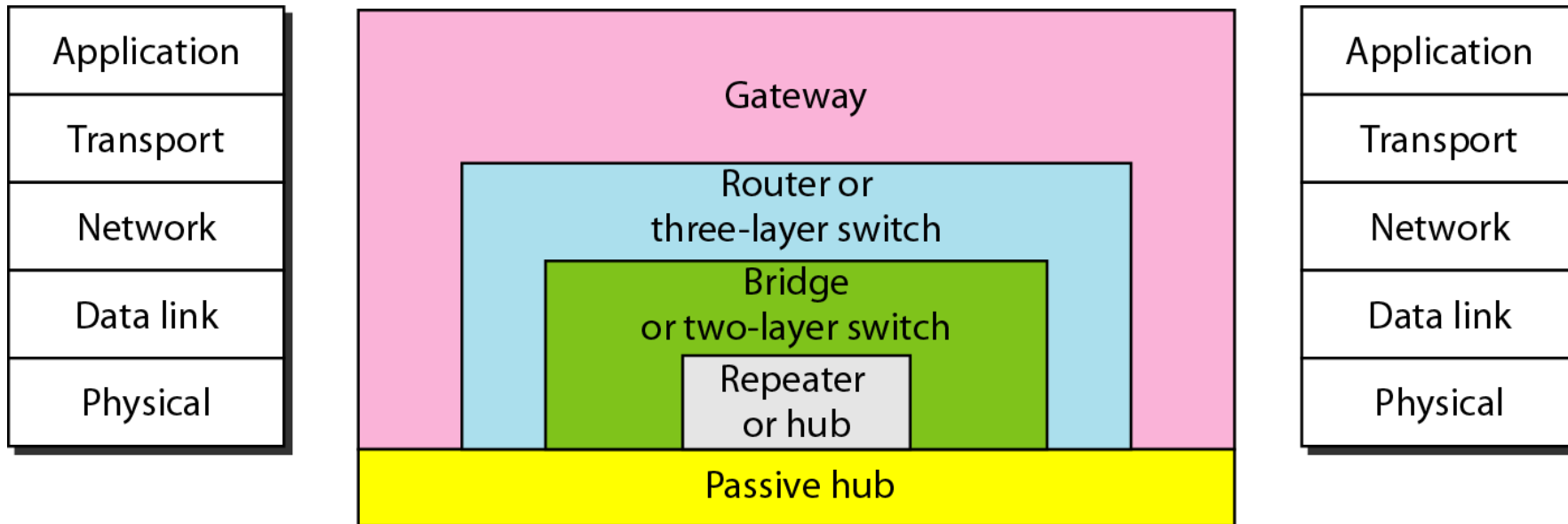
CLASSIFICATION OF NETWORKING DEVICES

Device	OSI Layer	Function
Repeater	Layer 1	Regenerates signals
Hub	Layer 1	Broadcasts data
Bridge	Layer 2	Filters frames
Switch	Layer 2	Forwards frames intelligently
Router	Layer 3	Routes packets
Gateway	Multiple Layers	Protocol conversion





CLASSIFICATION OF NETWORKING DEVICES





REPEATER

A **Repeater** is a Layer 1 networking device that **receives weak signals, regenerates them, and retransmits them** to extend the communication distance.

Characteristics

- Operates at Physical Layer
- No filtering
- No addressing
- Extends network distance

Advantages

- Increases transmission range
- Simple and inexpensive

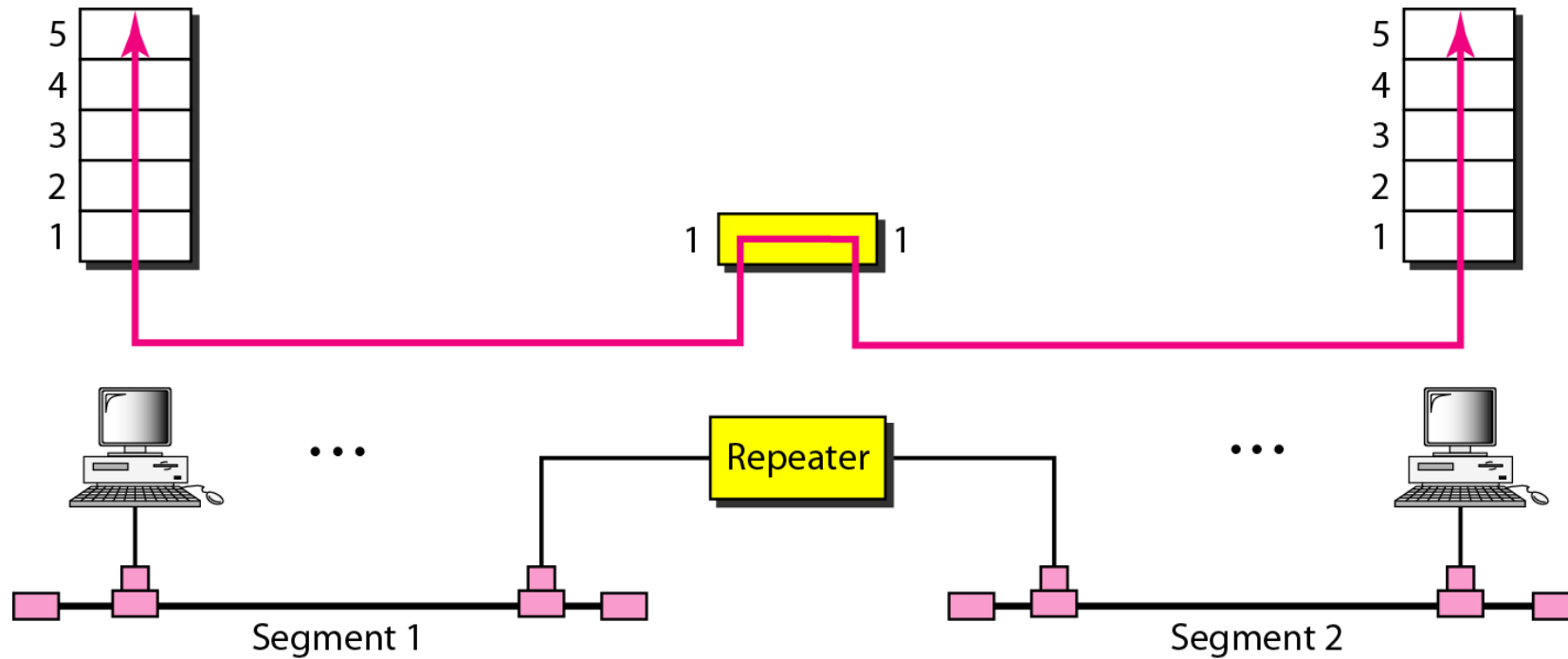
Limitations

- Cannot reduce network traffic
- Does not filter data

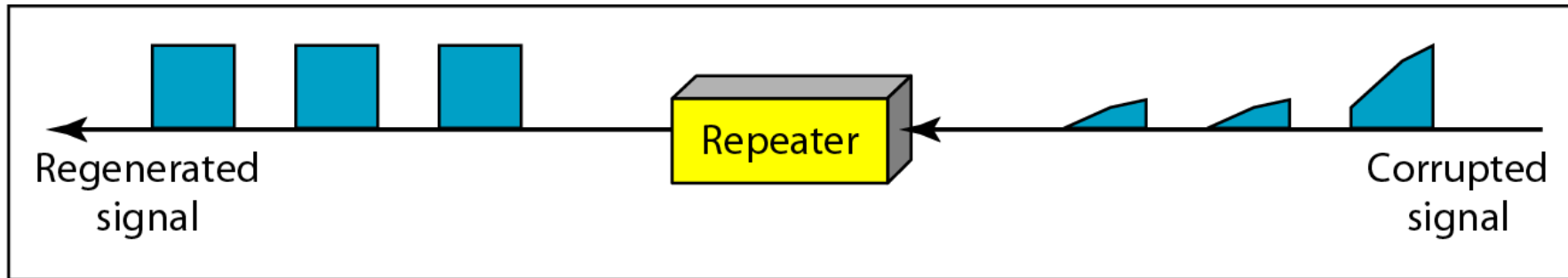
**A repeater is a regenerator,
not an amplifier.**



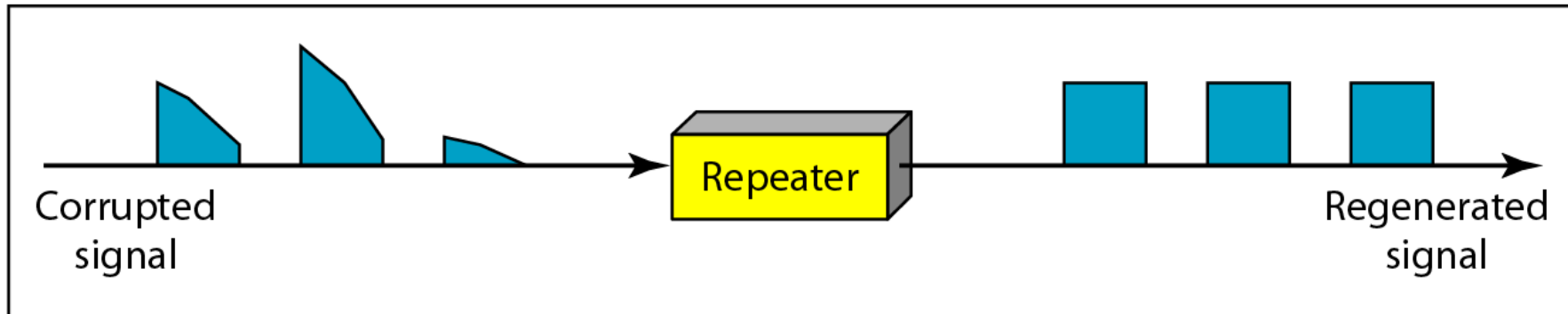
REPEATER



REPEATER



a. Right-to-left transmission.



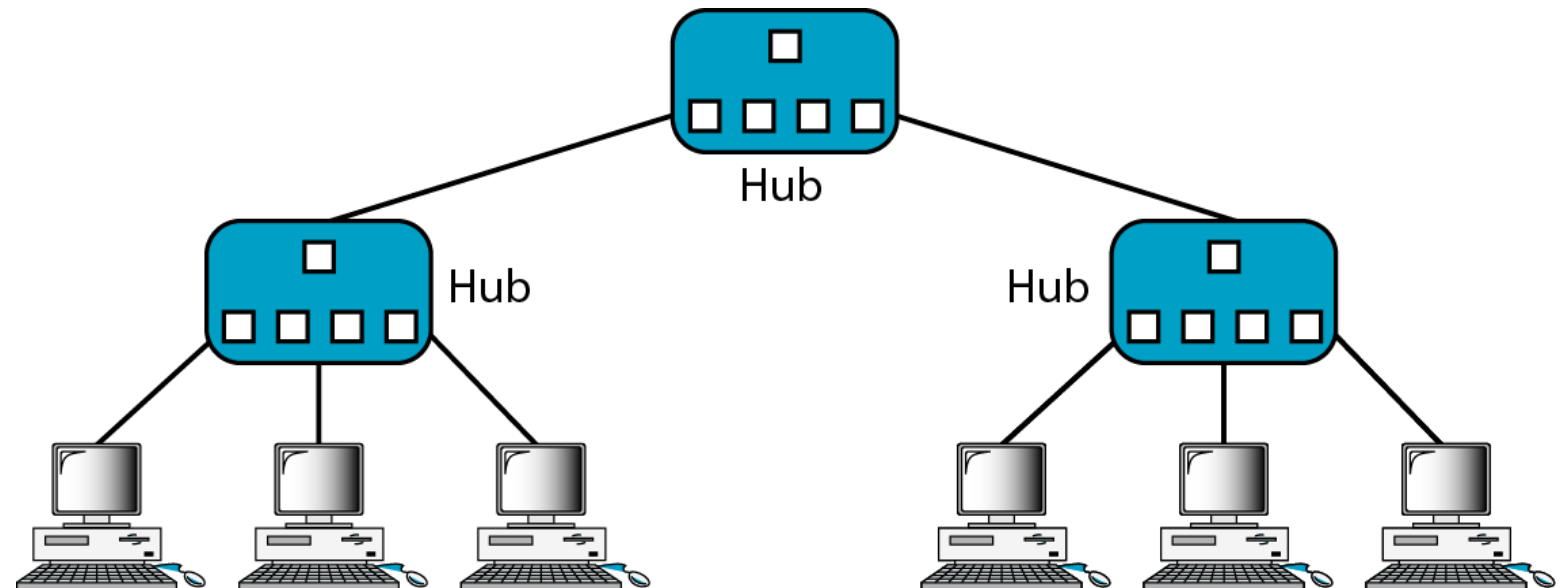
b. Left-to-right transmission.

HUB

*A **Hub** is a multiport repeater that connects multiple devices and broadcasts incoming data to **all connected devices**.*

Characteristics

- Physical Layer device
- Shared bandwidth
- Broadcast communication
- No MAC address learning
- Small LANs





BRIDGE

A **Bridge** is a Layer 2 device that connects two or more LAN segments and filters frames using **MAC addresses**.

Characteristics

- Data Link Layer device
- Learns MAC addresses
- Reduces collisions

Advantages

- Efficient traffic filtering
- Improves network performance

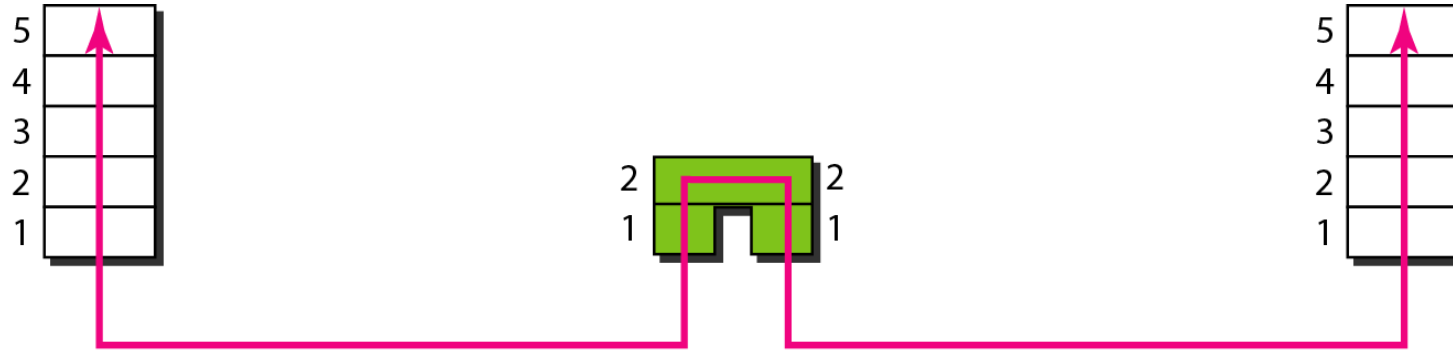
Applications

- Connecting LAN segments
- Office networks

A bridge has a table used in filtering decisions.

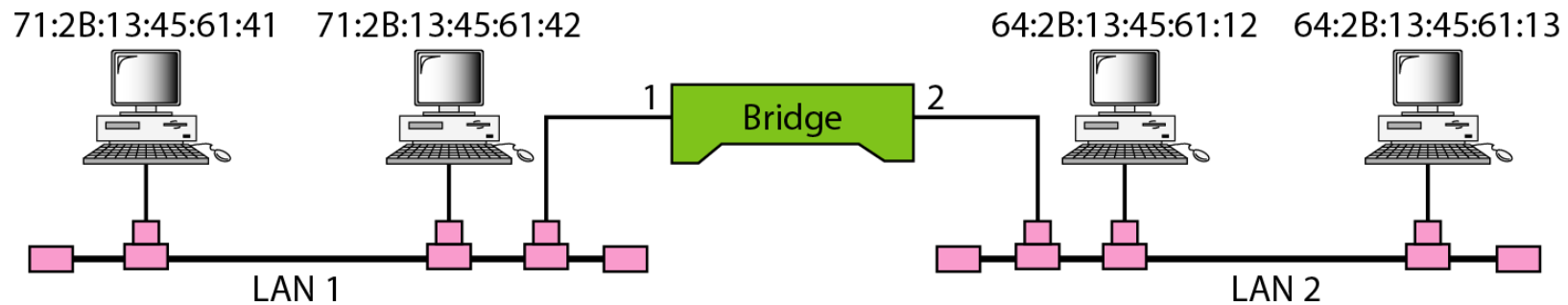


BRIDGE



Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table



BRIDGE

*A **Switch** is a Layer 2 networking device that forwards data frames **only to the intended destination** using a **MAC address table**.*

Characteristics

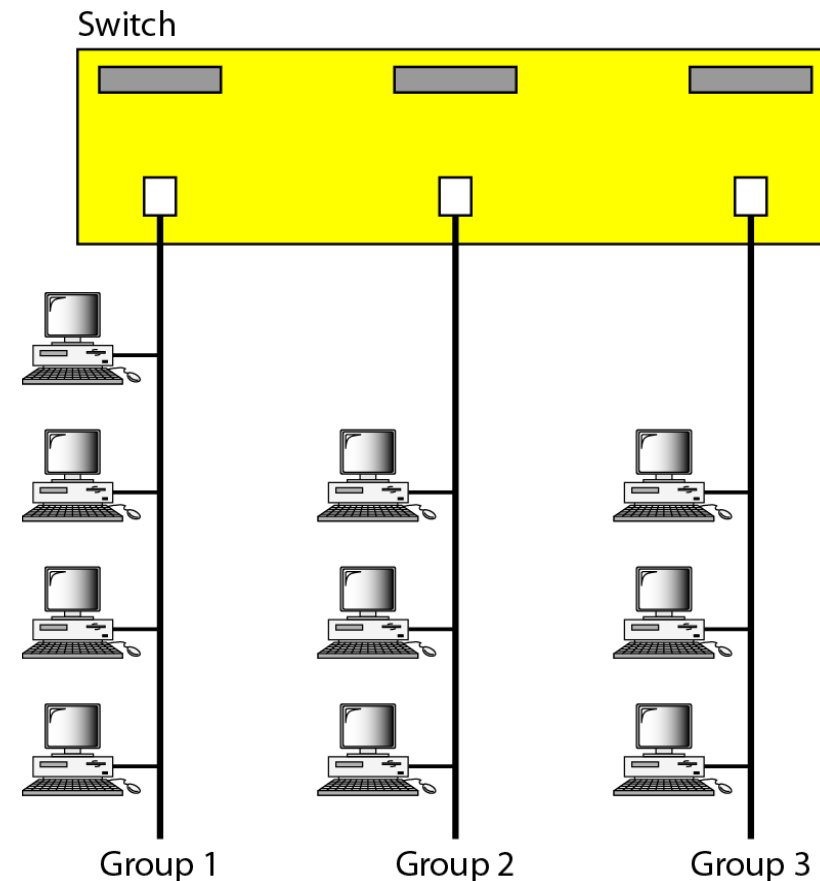
- Data Link Layer
- Intelligent forwarding
- Full duplex communication
- Dedicated bandwidth

Advantages

- Reduces collisions
- High performance
- Better security

Applications

- Modern LANs
- Campus Networks
- Enterprise Networks



ROUTER

*A **Router** is a Layer 3 networking device that connects different networks and forwards packets using **IP addresses**.*

Characteristics

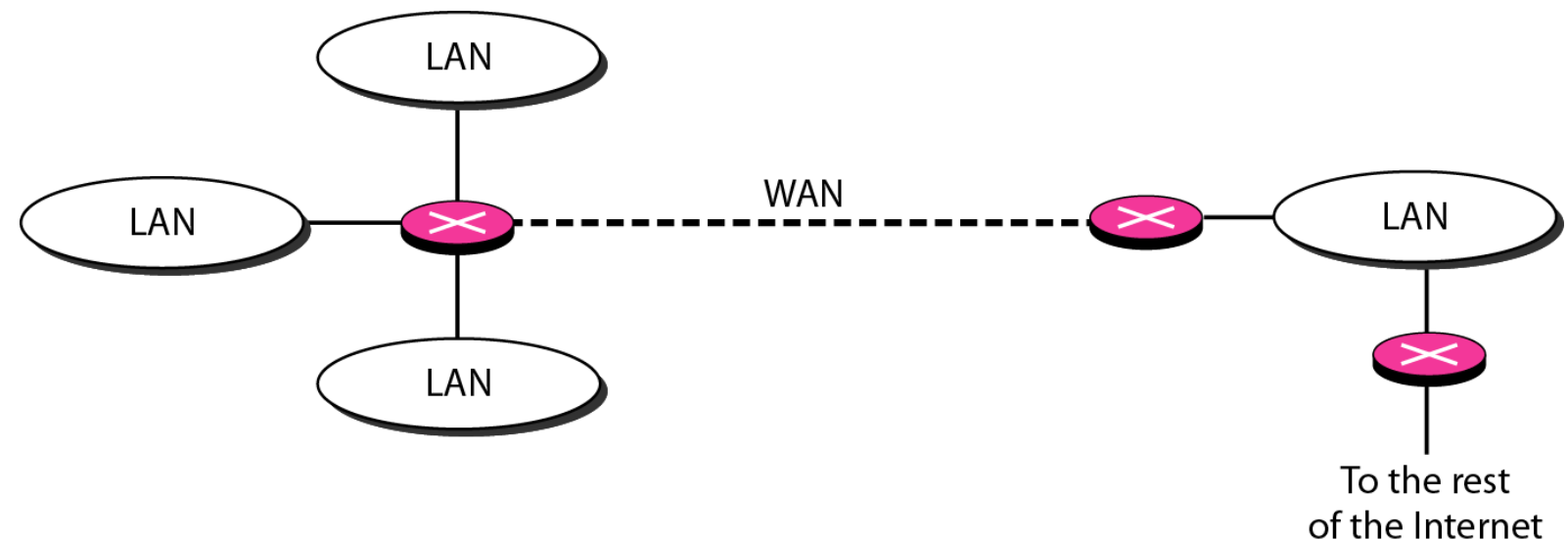
- Network Layer device
- Uses routing table
- Connects LAN to Internet

Advantages

- Connects multiple networks
- Chooses best path
- Reduces unnecessary traffic

Applications

- Home Wi-Fi Router
- ISP Networks
- Enterprise Networks



ROUTER

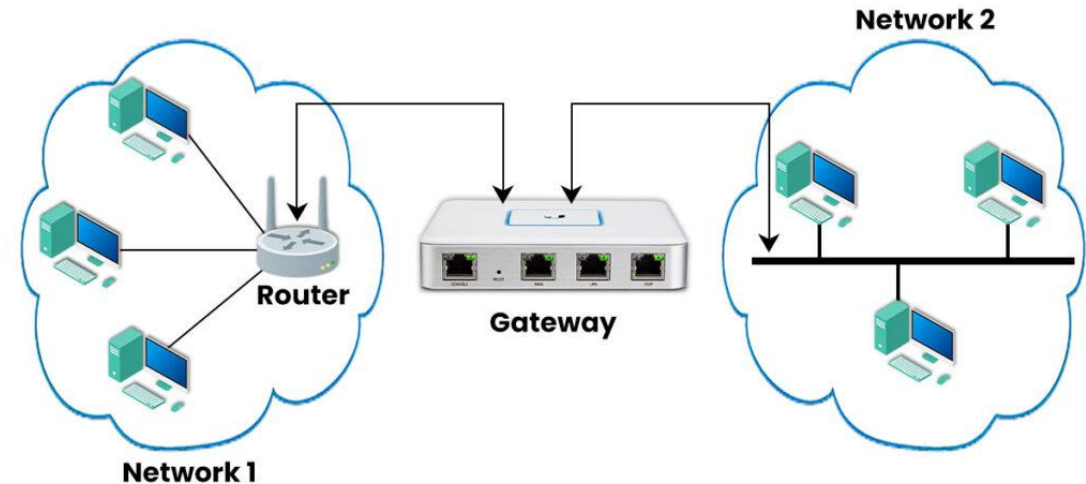
*A **Gateway** is a networking device that connects networks using **different communication protocols** by performing protocol conversion.*

Characteristics

- Operates at multiple OSI layers
- Protocol translation
- Data format conversion
- Enables communication between heterogeneous networks

Applications

- Email gateways
- VoIP gateways
- IoT gateways
- Cloud gateways





NETWORK INTERFACE CARD (NIC)

A Network Interface Card (NIC) is a hardware component that enables a computer or network-enabled device to connect to a network. It serves as the interface between the device and the transmission medium, allowing data to be transmitted and received.

Characteristics

Operates at the **Physical (Layer 1)** and **Data Link (Layer 2)** layers of the OSI model.

Connects a device to a wired or wireless network.

Sends and receives data over the network.

Provides a unique **MAC (Media Access Control) address** for device identification.

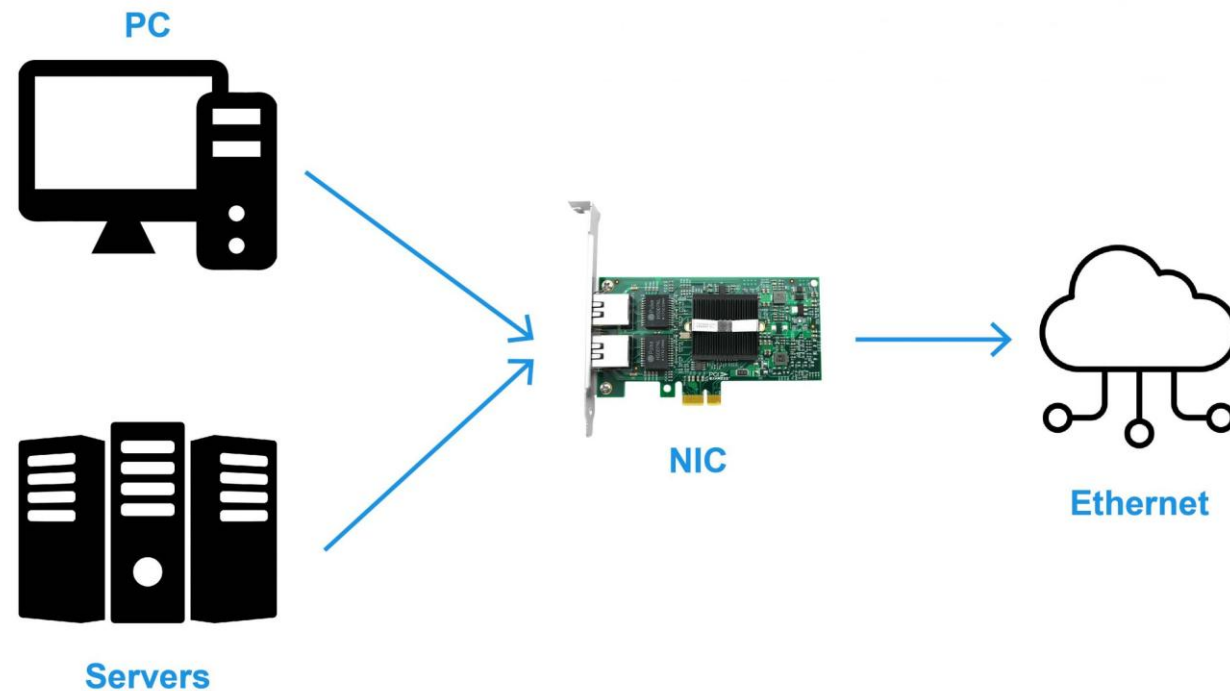
Types

Wired NIC (Ethernet Card) – Uses an Ethernet (RJ-45) cable.

Wireless NIC (Wi-Fi Adapter) – Uses wireless signals (Wi-Fi).



NETWORK INTERFACE CARD (NIC)





Thank you!

Doubts and Questions?